

Detection and Experimental Validation of High Impedance Arc Fault in Distribution System Using Empirical Mode Decomposition

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Abstract—The effect of high impedance faults (HIFs) in a distribution system has an effect on the system stability and reliability. The high impedance arc fault (HIAF) even poses a significant threat to the living being as it involves arcing. The HIAF have the same $V-I$ characteristics as HIF, but the enormous amount of heat generation is a significant concern to differentiate between them. Moreover, there are different types of arc which may occur depending on the arcing conditions and arcing surfaces. In this article, arc due to leaning-tree on a medium-voltage distribution line and the arc in between sphere-gaps are considered as the main event of analysis. An empirical mode decomposition (EMD)-based approach is used along with artificial neural network (ANN) for the analysis of real-time arc signals. The result obtained by the application of EMD and ANN on arc voltage signals successfully detects and classifies the arcing events by their predominant harmonic signatures. A comparison of EMD results with other techniques is presented. Arc in soil and wet-sand is also considered for the validation of the proposed algorithm for other arcing surfaces. Support vector machine and K th nearest neighbor machine learning techniques are also utilized for classification.

Index Terms—Empirical mode decomposition (EMD), fault detection, high impedance arc fault (HIAF).

NOMENCLATURE

ANN	Artificial neural network.
CV	Cross validation.
DWT	Discrete wavelet transform.
EMD	Empirical mode decomposition.
HHT	Hilbert–Huang transform.
HIAF	High impedance arc fault.
HIF	High impedance fault.
IMF	Intrinsic mode function.
KNN	K th nearest neighbor.
PSD	Power spectral density.
ST	Stockwell transform.
SVM	Support vector machine.
WT	Wavelet transform.
C_1	Intrinsic mode functions (IMF).
r_1 and r_2	Residue functions.
$m(t)$	Mean of the envelopes.

$s(t)$	Distorted signal.
τ	Time constant.
X_i	Whole dataset.
T_i	Training set.
V_i	Validation sets of different folds.

I. INTRODUCTION

THE HIAF occurs when the energized conductor makes an undesired contact with the ground surface, tree limb, or with any other surfaces. Moreover, it leaves the energized conductor open on the surface, which can pose a serious threat to the living beings. However, in practice, the conventional fault detection equipment fails to detect high impedance arc due to its minor impact on the current profile and almost all the equipment works on overcurrent principle [1], [2]. Moreover, the fault current may be less than one-tenth of the load current and seems to be not hurtful. However, it can certainly cause fire because of the heating associated with arc flash, which may rise to thousands of degree celsius over a long period. The most commonly occurred arc is between leaning tree and energized conductor. Due to storm in the monsoon, the tree branches sometimes lean toward the current carrying power conductors and cause an electrical arc. Then again, an electrical arc may also occur due to the fault in the equipment, insulation, or between two adjacent conductors. The fundamental difference of arc in leaning tree and electrical equipment is the grounding path for the fault current, which depends on grounding impedances. Moreover, the arcing surface is the primary factor for analyzing different arc faults. Therefore, the classification and detection of arc due to tree leaning is necessary to improve the protection scheme of a distribution system.

Over the years, many studies have been carried out on HIF, and experimentally observed that the randomness in the voltage and current waveform in the presence of HIF is due to the nonlinearity in fault impedance. However, many techniques such as power line carrier communication [3], mathematical morphology [4], [5], magnetic field signature-based analysis [6] are used to detect the HIF. On the other hand, harmonic analysis techniques [7] are also used for HIF detection. Scores of researchers have adopted DWT along with machine learning techniques [8], [9], such as the probabilistic neural network, ANN, SVM [10], etc., for the HIF detection. Choi–Williams distribution is also used along with time-frequency distribution for HIF detection [11], [12]. Apart from the simulations in [13] simulation as well as experimental verification of high

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impedance arc characteristics is described. In [13], a DWT-based detection technique is also proposed for arc due to the tree leaning on the power conductor. In [14], requirement for an HIF relay in distribution system and its details are discussed. HIF can provide a significant hazard in the coal mines as it can cause fire due to arc. A frequency characteristics-based detection technique for HIF is proposed in [15] considering dc trolley system of coal mines. However, the computational difficulty in frequency domain analyses motivated few researchers [16] to find the solution for arc detection even in time-domain. The phenomenon is also very much common in switchgear. In [17], a solution for arc flash mitigation in switchgear is proposed using high-speed switching. Several other researchers [18], [19] also have proposed different techniques to reduce the probability of arc hazard. The arc may occur in many places due to an equipment failure such as panel-boards, metal clad switchgear, switchboards, motor control centers, motor starters, and drive cabinets and transformers. These arcing events does not involve high impedance surface. Besides that, with the change of arcing surfaces the severity of the arc and the hazard associated with it also changes. Moreover, most of the literature reports analysis based on either current, voltage, or both measurement. The fault current during HIF becomes reasonably low, due to which conventional protection system does not respond to it. Therefore, a detailed analysis of arcing phenomena in a variety of arcing surfaces on the basis of arcing voltage is required and has not been addressed properly in the literature.

In this article, an approach based on EMD is adopted for the detection and classification of high impedance arc faults using arc voltage signals. The ANN, SVM, and KNN is also used along with EMD for machine learning. The arc signals are processed with EMD to get the decomposed signals, i.e., IMFs. These IMFs are further used as the inputs to the machine learning algorithms for classification. Furthermore, the outputs are comprehensively compared with the outputs of other signal processing techniques. The objective of this article is to choose appropriate signal processing and machine-learning tool for arc signals in order to analyze different arcing phenomena based on their harmonic signatures.

II. SIMULATION OF ARC IN DISTRIBUTION SYSTEMS

In a power distribution network, arcing phenomena is seen particularly in the time of natural calamities. Natural disasters can cause tree leaning in the distribution lines. However, causes of an arc are not limited to tree leaning. An arc is a short circuit through the dielectric breaking and could occur in a gaseous, solid, and liquid medium. In an arc flash, an ample measure of concentrated radiant energy detonates outward from electrical hardware in the form of a blast. The blast makes high-pressure waves that can pose a severe threat to the environment as well as to the living beings [1], [2]. Besides, arcing flash is the consequence of weak contact made between ground, a grounded object (solid, tree, etc.), and the conductor. Likewise, there might be an air-gap between line and ground. These air-gaps make a high potential for a short air-gap length, and the arc comes out when the air-gap brakes down. The fault current level amid the arc is insufficient to be recognized by the conventional fault-sensing equipment. One of the reasons

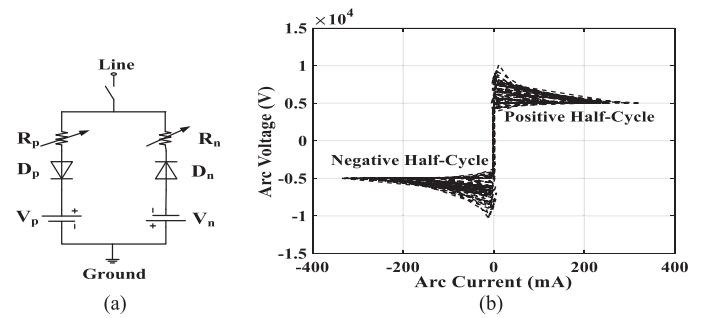


Fig. 1. Simulation model and result. (a) Arc simulation model. (b) Simulated arc characteristics due to a leaning tree.

is the continually varying condition of the contact surface. The surface, which is supporting the arc, also plays a vital role in the arc. Depending on contact surface, the arc impedance and conductance characteristics may change. Moreover, the surface impedance can be a linear or nonlinear one in nature based on the nature of the surface. The simulation of the arc is performed in a MATLAB environment. Two characteristics of an arc are simulated static and dynamic. The essential difference between both static and dynamic characteristics of an arc fault is in the time constant. For static, $\tau \approx 0$. Fig. 1(a) shows the arc fault simulation model. Fig. 1(b) describes the dynamic characteristics of the arc due to the leaning tree on the conductor. The impedance model presented in Fig. 1(a) controls the variation of impedance between ground and line [1], [2], [13], [20]–[22]. The arc characteristic where impedance does not change with respect to time is called static arc or static discharge. In Fig. 1(b), it is evident that characteristics are dynamic in nature due to time-dependent variation of impedances between the conductor and tree. In a practical situation, mainly dynamic arc is the most common type of arc.

Therefore, dynamic arcing has become a major area of research. When an arc occurs between a tree and live wire, the impedance changes frequently. As the current flows through a tree, it consistently is dried up, and its impedance continuously increases. However, this impedance variation does not have a fixed pattern. It is highly nonlinear in nature and produces a dynamic arc. In this article, different aspects of dynamic arcing phenomena are discussed.

III. MODIFIED HILBERT–HUANG TRANSFORM FOR ARC DETECTION

The Hilbert–Huang transform is a combination of Hilbert transform, which disintegrates transient signal into different intrinsic mode functions (IMFs) by applying EMD and the Huang transform, which extracts the instantaneous-frequencies and instantaneous-amplitudes of the IMFs. The subtle elements of the Hilbert–Huang technique alongside a few varieties and different enhancements are described in [23]–[25]. The intermittency signals and noises cause the problem of mode mixing [18], [19] in EMD. Furthermore, the technique of masking signals has also been suggested by few researchers [26], and it also can help EMD from mode-mixing. To construct appropriate masking signals to apply with the EMD the progressive FFT-based method has also been proposed [27]. Similarly, a demodulation approach

in postprocessing has also been proposed [28]. However, in reality, the amplitude from the Hilbert transform does not function as an envelope efficiently as the function has two spectrum lines [29], [30].

A. Empirical Mode Decomposition

The “derivative of the phase of an analytic signal” is portrayed as the fundamental approach of EMD [28], [31]. The application of the EMD is to distinguish different frequencies existent in time scales depicted by the interim between “local and global extrema’s.” A “local extremum” is defined by an instant when the first derivative of the signal is “zero,” but its second derivative is “nonzero.” The “global extrema” can also be obtained similarly. However, the “local and global extrema” are completely different from each other. For a case, inside an analysis window, there might be a few “local extrema.” However, just a single global extremum and a global minimum point might be available in a particular signal. In the process of EMD, once the time scales are recognized, IMFs with zero mean are extracted out of the analytic signal [27], [28]. Suppose, the distorted signal is “ $s(t)$ ” and the calculated IMF is “ $C(t)$.” If “ C_1 ” is an IMF, the number of “local extrema” of “ C_1 ” is equal to or differs from the number of zero crossings by one, and the average of C_1 is reasonably zero, then the residue can be calculated by

$$s(t) - C_1 = r_1 \quad (1)$$

$$r_1 - C_2 = r_2 \quad (2)$$

where C_1 , r_1 , r_2 stand for IMF and residue functions, respectively. So, the n th residue function will be

$$r_{n-1} - C_n = r_n. \quad (3)$$

Therefore, the final relation between all the IMFs and residue with the signal is as follows:

$$s(t) - \sum_{j=1}^n C_j = r_n. \quad (4)$$

The EMD finds out the envelopes $e_M(t)$ and $e_m(t)$, by performing cubic spline interpolation between maxima and the minima. If the mean of these envelopes is $m(t)$

$$m(t) = \frac{e_M(t) + e_m(t)}{2} \quad (5)$$

$$C_1 = s(t) - \sum_{j=1}^k m_j$$

$$\text{where, } r_1 = \sum_{j=1}^k m_j. \quad (6)$$

As we know from (2): $r_1 - C_2 = r_2$. Substituting the value of r_2 in (2) and combining with (6) we obtain the following:

$$C_2 = r_1 - r_2 = \left(\sum_{j=1}^k m_j - \sum_{j=1}^p m_{2j} \right). \quad (7)$$

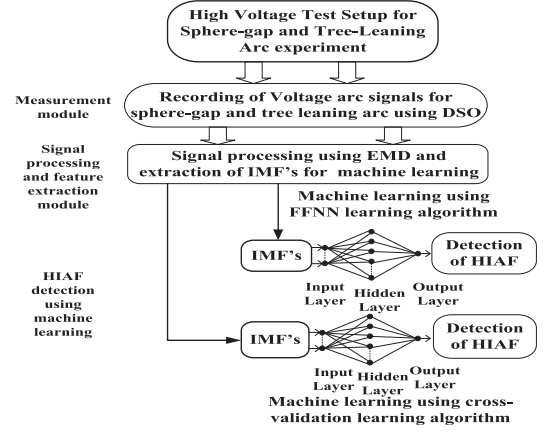


Fig. 2. Overall methodological representation of detection algorithm.

Adding the values of $c_1, c_2 \dots$ we obtain the following:

$$\sum_{j=1}^n C_j = s(t) - \sum_{j=1}^k m_j + \left(\sum_{j=1}^k m_j - \sum_{j=1}^p m_{2j} \right) + \dots = s(t).$$

Therefore, it concludes IMFs can reconstruct the main signal. However, the common practice is to avoid “overimproving” of $C_1(t)$ which causes a substantial loss of information [25]. The main advantage of this whole shifting is to eliminate the riding waves to make the number of extrema equals to that of zero-crossing and to make the envelopes symmetric.

B. Machine Learning Using Cross-Validation (CV) Algorithm

The easiest option to train and validate the network model would be the splitting of the dataset into multiple combinations of training and testing sets. This way of validating the data is commonly called CV algorithm. For example, if, $\{X_i T_i V_i\}$ is the whole dataset, training, and validation sets of different folds “ i ,” respectively. Here, the “fold” stands for the number of training and validation sets. The corresponding validation and training sets can be illustrated in the manner given as follows:

$$V_i = X_i \quad \text{and} \quad T_i = \bigcup_{i=1}^K X_i - X_i$$

where $i = 1, 2, \dots, K$. The final validation accuracy of the dataset is determined by taking the average of all the validation outputs of different folds. In this article, ten-fold CV is applied on the IMFs calculated from the fault signal for classification.

C. Fault Classification Using FFNN

Feedforward neural network (FFNN) is utilized for pattern recognition, calculating functions, and many other applications. This algorithm is a supervised learning algorithm. During the learning of FFNN, weights are adapted to optimize the network response. The different learning rule determines the weight updating process during learning. The most common algorithm for supervised learning in feed-forward networks is back-propagation [32]–[35] and the least mean square error algorithm. The whole methodological representation of the detection algorithm using FFNN and CV is shown in Fig. 2.

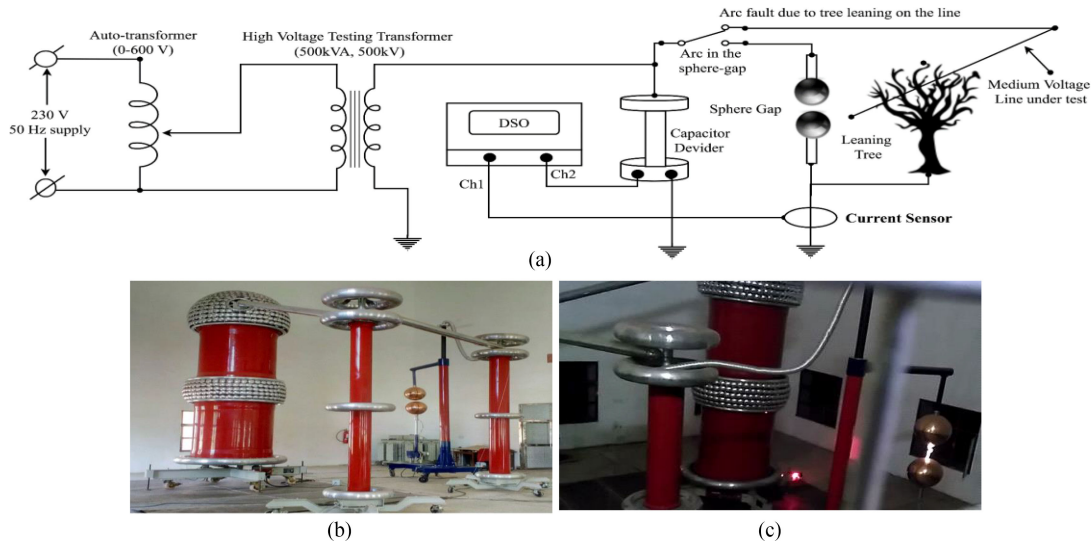


Fig. 3. Schematic diagram of the overall experimental setup and outcome for an arc in sphere-gap and leaning-tree. (a) Schematic diagram. (b) Experimental setup for the arc (1. HV testing transformer, 2. Capacitor divider, 3. Sphere-gap assembly, 4. Autotransformer) [37]. (c) Experimental setup during the arc (sphere-gap).

IV. EXPERIMENTAL ARRANGEMENT FOR ARC FAULT

It is challenging to simulate arcing phenomena in a real-time environment due to the equipment cost and the risk associated with it. In high-voltage engineering laboratory [36] at National Institute of Technology, Rourkela, India, a 500 kV, 500 kVA power frequency testing transformer (Make- W. S. Test Systems; Type-AC 500-500) is used for the experiment of arc fault both in sphere gap and tree-leaning conditions. It is a remotely operated HV laboratory with a capacity for experiments up to 500 kV level which includes extensive testing and detection of partial discharge (PD) with an advanced virtual instrumentation-based research facility, impulse, arcing, lightning phenomena, insulation failure, etc.

The rated I/O voltage for the testing transformer is 610 V/500 kV, 50 Hz and the rated test voltage is 450 kV. Fig. 3(a) explains the schematic diagram for sphere gap and tree leaning experimental arc arrangement. Here, an autotransformer is used as an input to the testing transformer. The autotransformer gives a voltage of 0–610 V to the low-voltage side of the HV testing transformer. The voltage from the testing transformer is applied to the sphere-gap to create an arc between them. Fig. 3(b) and (c) show the real experiment setup in HV laboratory. The applied voltage to all the arcing setup is measured through a potential transformer (PT) of 1000:1 ratio. The rated output power of the HV test setup is 500 kVA, 250 mA, (125 kVA). The detailed specification and descriptions are depicted in Table I. Moreover, it corresponds to the conducting surface with linear characteristics. On the other hand, the arc experiment is also performed for the tree-leaning on medium-voltage network. Due to storm or other natural calamity, very often the trees lean toward the medium-voltage network and causes arcing. In this case, the tree acts as an arcing surface with nonlinear high impedance. Fig. 4(a) shows an experimental arrangement of tree-leaning arc experiment. Fig. 4(b) shows arc in between tree leaves and the medium-voltage test line. Moreover, Fig. 4(c) shows the tree resistance variation in the laboratory environment for different voltages.

TABLE I
SPECIFICATION OF THE HIGH-VOLTAGE TEST AND MEASURING EQUIPMENT FOR ARC EXPERIMENT

Sl. No.	Description	Ratings
1	HV Test Transformer	Rated Test Voltage: 0–500 kV Rated output power: 500 kVA, 250 mA, (125 kVA) cont. duty 500 kVA, 1 A (500 kVA) 5 min ON; Rated PD level: less than 10 pC at rated voltage
2	Control Desk with oil cooled auto-transformer	320 kVA; Input: 2 Ph 415 V AC 50 Hz; Output: 0–610 V/420 A
3	Capacitor Measuring	500 kV, 1000 pF PD < 5 pC
4	Voltmeter	(AC peak- 4.5 Digit digital)
5	Compensating Reactor	0–600 V/50 Hz AC; 50/100/150/200 and 250 kVAR with common tapping C
6	Measuring Sphere-Gap	Diameter-500 mm

However, the main reason behind the nonlinear behavior of the tree during arc is the variation of conductance. Fig. 4(d) depicts the conductance variation of the tree during arcing with the input voltage. It is visible from Fig. 4(d) that conductance variation during tree-leaning conditions is more than sphere-gap and the nonlinearity associated with it is also visible during tree leaning arc. On the other hand, the air-gap between the sphere is more than the air-gap in leaning-tree arc experiment. However, similar level of arcing voltage is required for leaning tree due to the variation of arcing surface. In case of sphere-gap, the arcing surface is a conductor and the gap length is fixed. But, in case of leaning tree, the air-gap frequently changes and it offers a variable high impedance, which results in a dynamic arcing due to the change in the overall conductivity. Besides that, the disparity in the voltage transient overshoot is due to the overall conductance of the air-gap and arcing surfaces together. With the variation of water content in tree, the conductance also changes. Finally, it is clear that, due to the small air-gap in leaning-tree

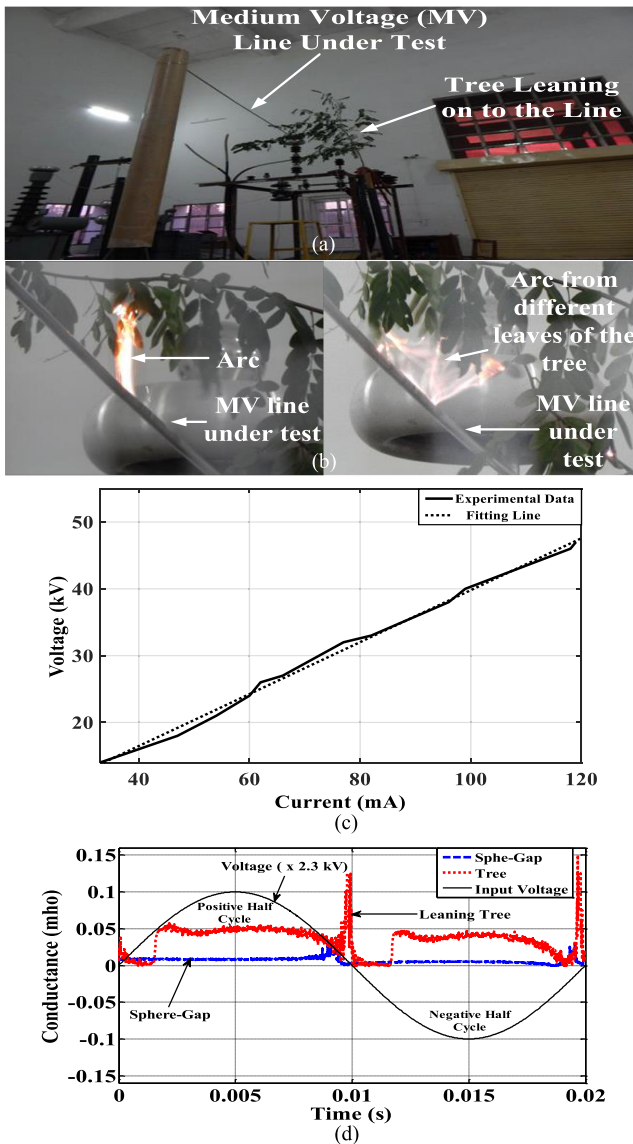


Fig. 4. Experimental setup for the arc between leaning tree and MV testing line. (a) Experimental setup for the arc between tree and line. (b) Arc from a single and different leaves of the tree during the experiment. (c) Tree resistance characteristics for different arc voltage. (d) Conductance variation of tree in comparison with sphere-gap during arcing along with input voltage.

experiment, the overall conductance is always higher than the sphere-gap. On the other hand, the conductance variation in wet-sand and soil are also very much different from each other due to its kind. It is due to the type of surface and the presence of water in it. Unlike wet sand, soil mounds create irregularity in the current conducting path. Therefore, the overall conductance variation in any surface varies with its kind.

V. EXPERIMENTAL RESULTS AND ANALYSIS

The $V-I$ characteristics of arc fault can vary depending on the nature of arc. The time constant of an arc determines the $V-I$ characteristics. On the other hand, several arc characteristics may also change depending on other parameters. In this article, two types of arc are seen after two different sets of experiments for the analysis. In the first experiment, the arc is created between

sphere-gap. Whereas, in the second experiment, the arc is created between a medium-voltage line and leaning tree. In both the cases, the arc voltage and arc current are captured through PT and current sensors. The real-time signals and the $V-I$ characteristics of both sphere-gap arc and tree leaning conditions are recorded through the oscilloscope. Fig. 5(a) and (b) represent the voltage, current signals, and corresponding $V-I$ characteristics for the arc in sphere-gap and leaning tree respectively. Fig. 5(c) and (d) describe the voltage and current transients due to the arc for both tree and sphere-gap arcing. Though the $V-I$ characteristics of both the arcing events are similar to each other, it is evident from Fig. 5(c) and (d) that it brings on a different type of transients. In the case of tree-leaning, the voltage and current transient during arc are attaining much higher amplitude than the sphere-gap. Sometimes the voltage overshoot is reaching beyond 50 kV for tree-leaning condition, whereas, in sphere-gap, the overshoot is around 15–20 kV. These variations are visible from the zoomed figures of Fig. 5(c) and (d). The experimental results presented in Fig. 5(a) and (b) are with voltage level 39 and 36 kV, respectively. The sampling rate for the acquisition of the voltage and the current signal is 500 kHz, and all the processing is performed on a PC with Intel Core i7-3770 processor, which has a processor speed of 3.40 GHz. The $V-I$ characteristics for both the cases are very much identical to the standard arc characteristics. Moreover, the time domain response in both the cases resembles with HIF.

However, the difference in the two type of arc is also prominent from the nonlinear variation of conductance during tree-leaning arc experiment depicted in Fig. 4(d). On the other hand, in the case of tree-leaning, the voltage and current transient during arc are attaining much higher amplitude than sphere-gap arcing. Sometimes the voltage overshoot is reaching beyond 50 kV for tree-leaning condition, whereas, in sphere-gap, the overshoot is around 15–20 kV.

A. Frequency-Based Detection of Arc Signals Using EMD

In this article, the EMD is applied on the voltage signals. Fig. 6(a) and (b) represent the decomposed IMFs and their PSD plots for the voltage signals, respectively. Fig. 6(b) also informs the predominant frequency with maximum spectral density for each IMF. Total ten (IMF 1–IMF 10) number of IMF has been calculated after applying EMD from the voltage signal. The analysis of IMFs and their PSD plots describe the presence of two predominant fundamental frequencies (shown as 49.9 Hz) with different power spectral density ([IMF 8 and IMF 9 of Fig. 6(b)] along with other higher order predominant frequencies in other IMFs. However, the predominant frequency calculation is performed only to describe the frequency content and the PSD distribution among the frequencies present in the IMFs. On the other hand, the total number of IMFs calculated from the EMD of arc voltage signal for leaning tree is 11 (IMF 1–IMF 11). Fig. 5(b) shows current and the voltage waveform of arc due to the tree leaning on MV line, respectively. The EMD response from the voltage signal and the PSD plots are shown in Fig. 7(a) and (b) respectively. In addition to that, Fig. 6(b) and (b) also depict the predominant frequency with maximum spectral density for each IMF. Unlike arc in sphere-gap, the tree leaning arc gives three predominant fundamental frequencies

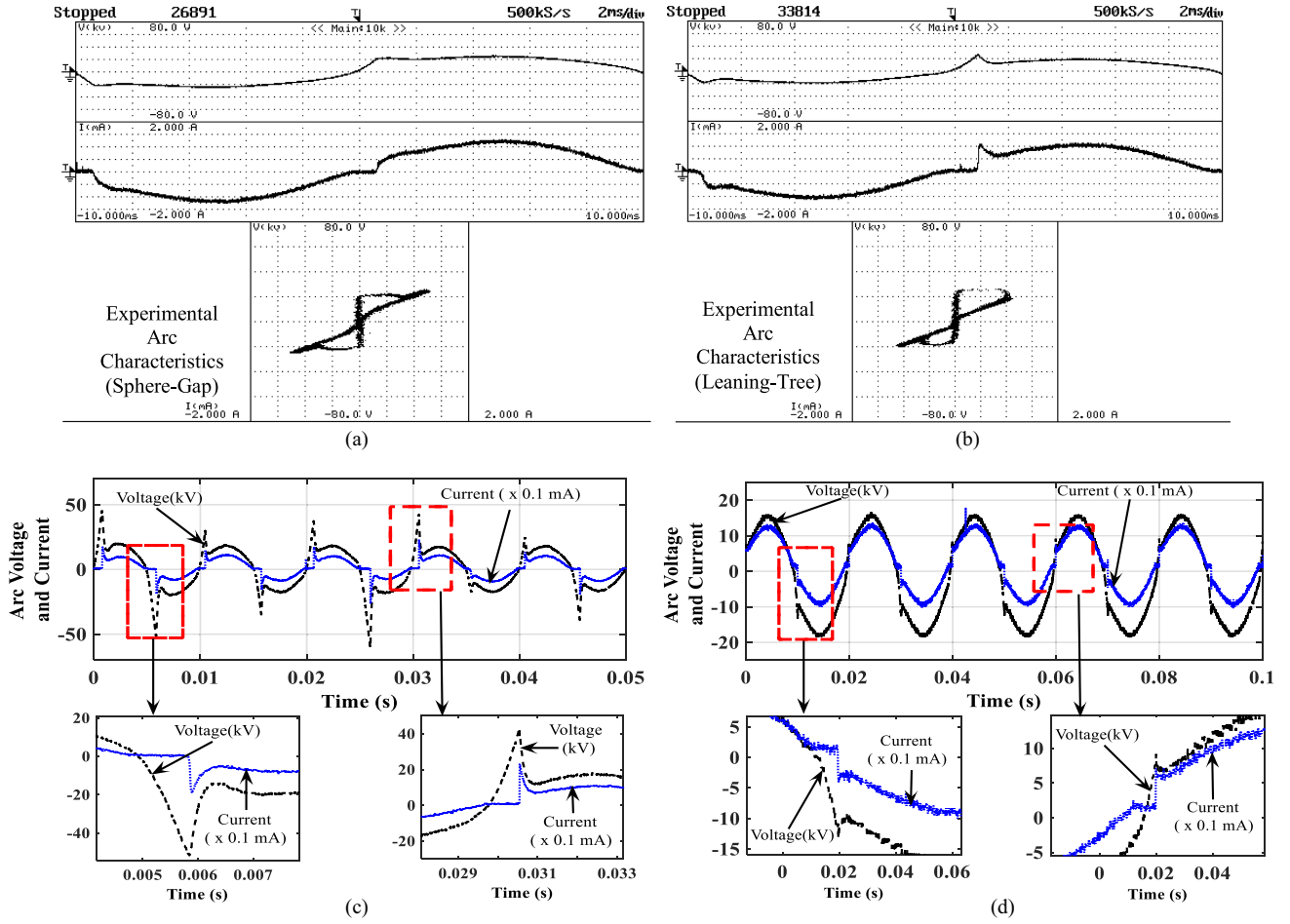


Fig. 5. Experimental result for arc in sphere-gap and leaning tree captured through the voltage and current sensors. (a) Experimental result for arc between sphere-gap at 39 kV. (b) Experimental result for arc between tree leaf and MV line at 36 kV. (c) Combined voltage and current signal for the tree-leaning arc. (d) Combined voltage and current signal for the sphere-gap arc experiment results.

TABLE II
PREDOMINANT FREQUENCIES OF IMFs CALCULATED FROM TREE-LEANING
AND SPHERE-GAP ARCING VOLTAGE

IMFs	Predominant frequency (Hz)	
	Sphere-Gap	Leaning-Tree
IMF 1	18962.07	20209.58
IMF 2	18762.47	69361.28
IMF 3	34730.54	28942.11
IMF 4	21007.98	17514.97
IMF 5	6736.53	9381.24
IMF 6	349.30	748.5
IMF 7	598.8	49.90
IMF 8	49.9	99.80
IMF 9	49.9	49.90
IMF 10	0	49.90
IMF 11	Not Found	0

(49.9 Hz) component with different PSDs [see IMF 7, IMF 9, and IMF 10 of Fig. 7(b)] with other higher order frequencies. On the other hand, it is also observed from the frequency response that, the second-order harmonic is predominant [see IMF 8 of Fig. 7(b)] in the frequency spectrum due to the maximum spectral density in the case of tree, which in the event of sphere-gap is not predominant. Table II shows the predominant frequencies

of IMFs calculated from tree-leaning and sphere-gap arcing voltages. Table II also shows the basic differences between two different arcing scenarios and some similarities regarding their frequency information. The similarities between both the experiments are the presence of more than one IMF of having a predominant frequency of 49.9 Hz (fundamental frequency) due to the nonlinear behavior of voltage during both the arcing. Therefore, it is clear that both the arcing phenomena is having a nonlinear behavior.

Furthermore, the main difference lies in the presence of 99.8 Hz (second-order harmonics) in case of the only tree-leaning arc, which makes it resemble perfectly with the other HIF conditions for both V-I characteristics and frequency response. Furthermore, the predominant frequency in IMF 10 of sphere-gap and IMF 11 of tree-leaning arc experiment is found to be "0," because the dc component has the highest PSD in those IMFs. Furthermore, IMF 11 is not found in case of sphere-gap arcing voltage, as the residue becomes zero after extracting ten IMFs. Therefore, by harmonic analysis with EMD, the HIAF is detectable among different arcs. In comparison with other HIFs, the HIAF due to the tree leaning on MV line shows similar characteristics which sphere-gap arcs are not showing regarding frequency signature.

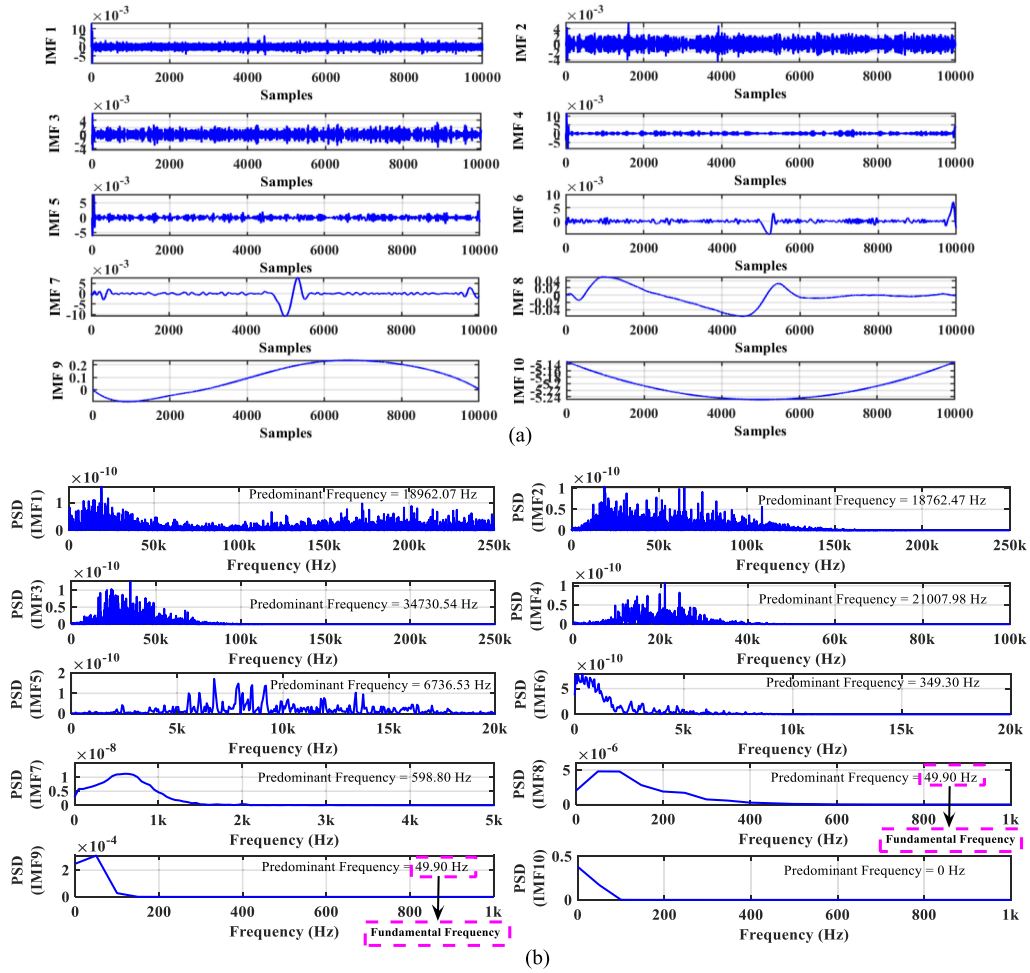


Fig. 6. Testing results for sphere-gap arcing at 39 kV arcing voltage. (a) EMD response or IMFs from the voltage signal. (b) Power spectral density plots of the IMFs obtained from voltage signal.

B. Arc Fault Classification Using EMD-ANN Approach

The frequency response of the arc signals using EMD is giving a shred of evidence to differentiate between sphere-gap arcing and leaning tree arcing. To validate the observations and results, a machine learning algorithm is applied with proper feature selection technique to classify two different faults. In this scenario, FFNN and ten-fold CV is used as a machine learning tool to check the validity of the EMD analysis claims. Fig. 2 presents the overall methodological representation of HIAF detection using FFNN and CV machine learning algorithms. The results for both the learning algorithm FFNN and CV are recorded for the cross verification. As the number of IMFs are not same (tree:11; sphere:10) for both the cases, the maximum number of IMFs considered for the machine learning is 10. The neural networks are trained for two different set of inputs such as ten IMFs (tree: IMF 2–11; sphere: IMF 1–10) and five IMFs (tree: IMF 7–11; sphere: IMF 6–10). Fig. 8(a) and (b) show the training regression plot for ten and five IMFs, respectively, and Fig. 8(c) and (d) depict the training performance plot for ten and five IMFs, respectively. The regression plots and the training performance plots are supporting the initial claims of fault classification using EMD alone for both sphere-gap and tree leaning arcing.

TABLE III
CLASSIFICATION RESULTS USING FFNN AND CROSS-VALIDATION-BASED APPROACH

Number of Features	Selected Features	Classification Accuracy (%)	
		CV	FFNN
Tree-11	10 (IMF 2-11)	99.99	99.97 (Network model: 5:8:1)
Sphere-gap-10	10 (IMF 1-10)		
Tree-11	10 (IMF 7-11)	99.67	99.97 (Network model: 5:8:1)
Sphere-gap-10	10 (IMF 6-10)		

Table III illustrates the results recorded after both the learning algorithms are applied with the same kind of inputs. The results are not contradicting each other and unanimously supporting the outcomes and claims of EMD-based analysis. Therefore, the voltage and current profile of the tree leaning condition is more distorted than the sphere-gap condition. However, the main reason behind the two different responses of arc is the added nonlinear impedance due to the tree and the variation of impedance path due to the burning of leaves on the course of arcing. In the case of HIF, the presence of second harmonics is the most prominent feature to have resembled as HIF regarding

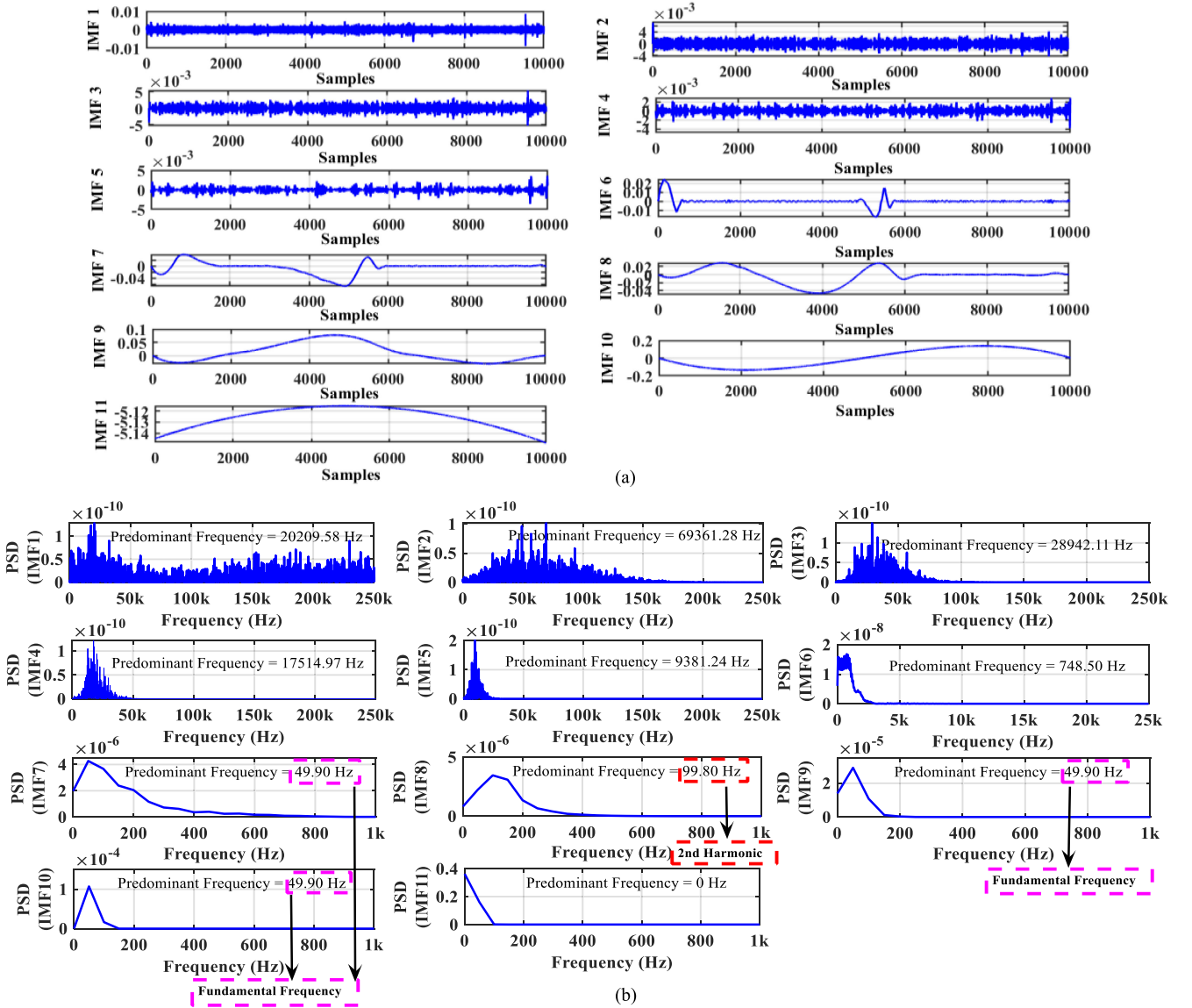


Fig. 7. Testing results for tree-leaning arcing at 33 kV arcing voltage after applying EMD on voltage signals. (a) EMD IMFs calculated from the voltage signal. (b) Power spectral density plots of the IMFs obtained from voltage signal.

frequency response [20]. Moreover, this feature plays an important role while classifying HIF and low-impedance fault. The experiments are repeated and performed with different voltage levels, and every time it validates the results presented in this article.

Although the arc in sphere-gap resembles with other HIFs regarding $V-I$ characteristics, it differs from the very basic idea of HIF concerning frequency signature. Moreover, the training results are also compared with the results obtained from the coefficients applying different wavelet transforms using different mother wavelets considering CV analysis. Table IV presents a comparative picture of the results obtained with different signal processing techniques like Gaussian wavelet transform (GWT), continuous wavelet transform (CWT) with different mother wavelets. As the second harmonic can be the basis of the analysis, the scales and mother wavelets are selected accordingly. The classification accuracies obtained with different

TABLE IV
COMPARATIVE ANALYSIS BETWEEN EMD AND WAVELET

Signal Processing Tools	Number of Features	Selected Features	Classification Accuracy (CV)
EMD	Tree-11 Sphere-gap-10	10 (IMF 2-11) 10 (IMF 1-10)	99.99 %
GWT	10	10	63.26 %
CWT ('Coiflet 3')	10	Coefficient for 10 scales (349 to 358)	53.34 %
CWT ('Db 4')	10	Coefficient for 10 scales (352 to 361)	52.08 %
CWT ('Db 8')	10	Coefficient for 10 scales (352 to 361)	53.95 %

signal processing techniques clearly shows the superiority of EMD over other signal processing techniques.

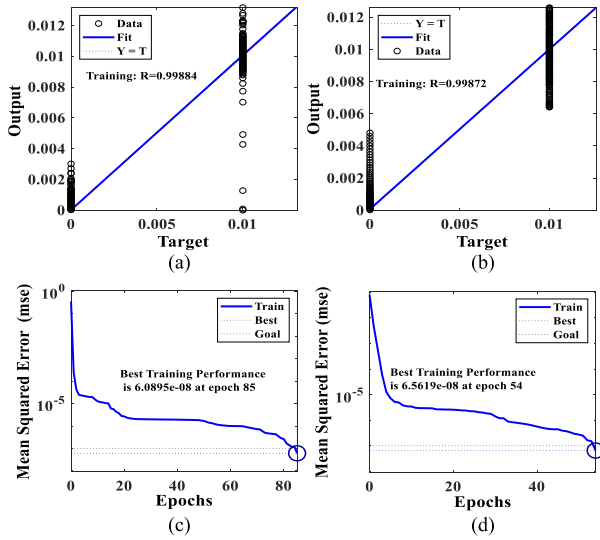


Fig. 8. Training performance and regression plot using EMD-ANN-based approach for sphere-gap and tree leaning arc fault classification. (a) Regression plot considering ten IMF as input feature. (b) Regression plot considering five IMF as input feature. (c) Performance plot considering ten IMF as input feature. (d) Performance plot considering five IMF as input feature.

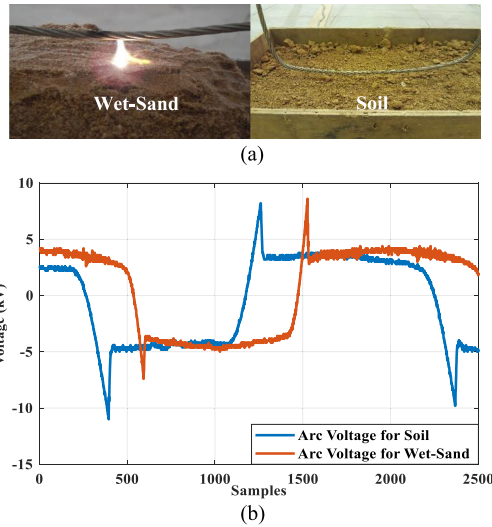


Fig. 9. Arc in wet-sand and soil. (a) Experiment setup for arc in wet-sand and soil. (b) Voltage profile of arc in soil and wet-sand.

C. HIAF Detection in Soil and Wet Sand Using EMD for Broken Conductor

To study the behavior of HIAF with the variation of arcing surface, two more surfaces are considered for analysis. In the first experiment, the arc is created between the sphere-gap and leaning-tree. Whereas, in the second set of experiments, the arc is created in wet-sand and soil. The experiment photos for both the events are presented in Fig. 9(a). In both cases, the arc voltage and arc current are captured through PT and current sensors like before. In the case of wet-sand, the stress created due to the air-gap between sand and the broken conductor displaces sand and finds the low impedance path possible. On the other hand, the different thickness of soil bed is considered. However, the

TABLE V
ARC CLASSIFICATION BETWEEN SOIL AND WET-SAND
USING EMD AND DIFFERENT SVM

Training set		Signal Processing Tool	Machine Learning Tool (Different SVM)	Classification Accuracy (CV)
Soil Arc Voltage (kV)	Wet-Sand Arc Voltage (kV)			
25	17	EMD	Quadratic	100 %
			Cubic	100 %
			Fine Gaussian	99.1 %
			Medium Gaussian	99.8 %

TABLE VI
ARC CLASSIFICATION BETWEEN SOIL AND WET-SAND USING EMD AND DIFFERENT KNN

Training set		Signal Processing Tool	Machine Learning Tool (Different KNN)	Classification Accuracy (CV)
Soil Arc Voltage (kV)	Wet-Sand Arc Voltage (kV)			
25	17	EMD	Weighted	99.3 %
			Cubic	98.2 %
			Fine	99.7 %
			Medium	98.7 %

difference was with the current. The reason behind the difference is the difference in the offered fault impedances of tree, wet-sand, and soil. The voltage waveform for arc in wet-sand and soil is shown in Fig. 9(b). The real-time signals are captured using PT and sensors. As the EMD is suitable for arc signals, IMFs for the soil and wet-sand arc voltages are calculated.

The IMF plots and their corresponding PSD plots are shown in Fig. 10. The PSD plots for both the events depict the presence of even-order harmonics (fourth) and multiple number of fundamental frequency with different power spectral density. The reason behind the presence of even-order harmonics being predominant in the IMF is the high impedance arcing surfaces. These IMFs are further used for the classification of arc voltages using different SVM and KNN. The classification results for SVMs and KNNs are presented in Tables V and VI, respectively. The number of IMFs which is considered for the analysis is 8. Table VII presents more precise details of probabilities of missed and false detection for both KNN and SVM. The results obtained after machine learning clearly shows that the arc voltages are successfully classified and the performance of KNN and SVM in this regard is consistent. Therefore, it is evident that the characteristics and consequences of different arcing conditions are very much different from each other. However, the difference between the two types of arc is also prominent from the nonlinear variation of voltage during different experiments. Nevertheless, the overall difference and the importance of four separate experiments are the grounding path for the discharge current. However, the time-domain responses of different arc faults are not very contrary to each other.

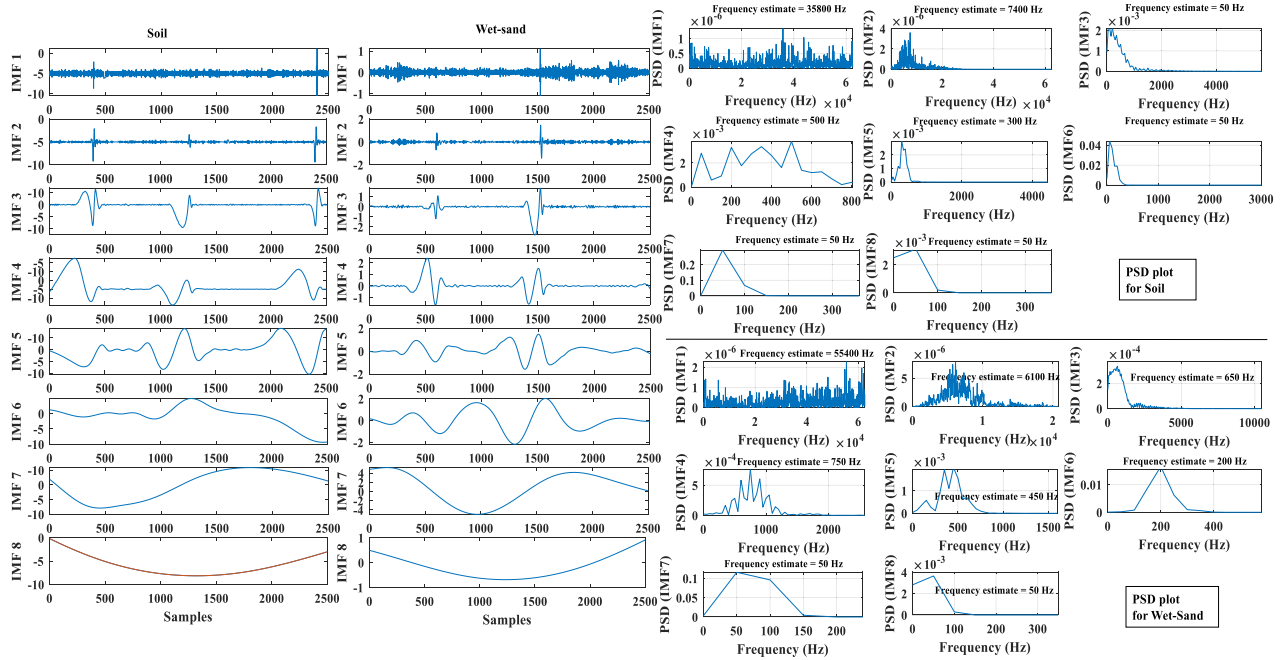


Fig. 10. IMFs and their corresponding PSD plots for soil and wet-sand arcing at after applying EMD on voltage signals.

TABLE VII
PROBABILITY OF MISSED AND FALSE DETECTION FOR KNN AND SVM

Machine Learning Tool	Classification Accuracy (CV)	Probability of Missed Detections (%)	Probability of False Detections (%)
Weighted KNN	99.3 %	<1	1
Cubic KNN	98.2 %	1	3
Fine KNN	99.7 %	<1	<1
Medium KNN	98.7 %	<1	2
Quadratic SVM	100 %	0	0
Cubic SVM	100 %	0	0
Fine Gaussian SVM	99.1 %	0	2
Medium Gaussian SVM	99.8 %	0	<1

TABLE VIII
COMPARATIVE ANALYSIS WITH GHADERI *et al.* [12]

Signal Processing Technique	Detection tool	Classification Accuracy (CV)
Ghaderi et.al. [12] {Measurements: Time-Domain, Frequency domain etc. Current, Current and Voltage}	Threshold-counter, SVM, ANN, decision tree, fuzzy-based classifier etc.	Highest: 100% Lowest: 80%
Present Study {Voltage}	EMD FFNN, SVM(quadratic, cubic) and KNN	Highest: 100% Lowest: 98.2%

D. Comparative Study

In [12], a review has been done based on HIF detection and localization techniques. An overall qualitative and quantitative comparison is provided based on overall accuracy, dependability, security, etc. It gives a clear picture of the reported literature in this field of research. However, the basic difference between the present article and the reported literature is the basis of analysis. There is no accuracy comparison reported using the arc voltage as the basis of analysis. On the other hand, most of the literature are classifying between no-fault and HIF. The present article focuses on the analysis based on the voltage measurement. The current during HIF becomes reasonably low, due to which conventional protection system does not respond to it. A comparative study with [12] is presented in Table VIII. Moreover, it also classifies between different HIFs depending on different arcing surface.

VI. CONCLUSION

The results obtained from both the experiments of sphere-gap and leaning tree clearly show the differences in the time-frequency domain analysis. However, the $V-I$ characteristics of both the event are quite identical and show similar behavior as HIF. But, the time-frequency domain analysis using the EMD method or modified Hilbert–Huang transform successfully classifies the two events by second-order harmonics. In case of tree leaning experiment, the presence of second-order harmonics is quite similar to HIF which the sphere-gap arc does not show. For soil and wet-sand experiment instead of second-order harmonics, fourth-order harmonics has been found as a signature of HIF. The results obtained from the real-time signals using EMD also confirms the presence of fundamental frequency with more than one PSD, which is also an important feature of the HIF due to the presence of nonlinearity in fault impedances. Moreover, the main difference between the four testing condition is arcing

surface. In the case of sphere-gap, the arcing surface is a metallic surface which offers linear impedance. However, in case of the tree-leaning, soil, and wet-sand experiment, the arcing surface offers nonlinear high impedance. The main conclusions of the article are as follows.

- 1) High impedance arc fault can be of different types depending upon the grounding path or the arcing surface.
- 2) HIF can be detected from the arc voltage signals using EMD.
- 3) In HIAF, fundamental frequency can be found to be more than one PSD value, and it also changes with the arcing surface.
- 4) Unlike HIAF, the sphere-gap arc may not contain predominant second-order or even-order harmonics.
- 5) The classification results using FFNN, SVM, and KNN learning algorithms unanimously support the outcome of EMD analysis by classifying all the events with a good level of accuracy.
- 6) As compared to other signal processing techniques, EMD is more suitable for the analysis of arc signals.
- 7) The potential of KNN and SVM is also found to be promising for arc classification.

This article is a small step toward the development of signal processing-based fault diagnosis system. The future research is planned for different type of nonlinearity offered by various surfaces.

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